MH3600 Topology and Manifolds

Solutions of Tutorial #8

Topics covered:

- Compactness
- Manifolds

Problem 1

(a) Let $\mathcal{T}$ and $\mathcal{T}'$ be two topologies on the set X; suppose that $\mathcal{T}' \supset \mathcal{T}$. What does compactness of X under one of these topologies imply about compactness under the other?

(b) Show that if X is compact Hausdorff under both $\mathcal{T}$ and $\mathcal{T}'$, then either $\mathcal{T}$ and $\mathcal{T}'$ are equal or they are not comparable.

Solution: **(a)** Compactness in $\mathcal{T}'$ implies compactness in $\mathcal{T}$, since every open cover in $\mathcal{T}$ is an open cover in $\mathcal{T}'$. The reverse implication is not true, consider $[0,1] \subset \mathbb{R}$ in the usual topology and the discrete topology.

(b) Suppose $\mathcal{T}$ and $\mathcal{T}'$ are comparable. Without loss of generality suppose $\mathcal{T} \subset \mathcal{T}'$. We will show that $\mathcal{T}' \subset \mathcal{T}$. Let $U \in \mathcal{T}'$. Then $X - U$ is closed in the $\mathcal{T}'$ topology. Since X is compact, Theorem 26.2 tells us that $X - U$ is also compact in the $\mathcal{T}'$ topology. Then by part (a), $X - U$ is compact in the $\mathcal{T}$ topology. Since X is Hausdorff and $X - U$ is compact, by Theorem 26.3 tells us that $X - U$ is closed in the $\mathcal{T}$ topology. Thus $U \in \mathcal{T}$. $\blacksquare$

Problem 2

(a) Show that in the finite complement topology on $\mathbb{R}$, every subspace is compact.

(b) If $\mathbb{R}$ has the topology consisting of all sets A such that $\mathbb{R} - A$ is either countable or all of $\mathbb{R}$, is $[0,1]$ a compact subspace?

Solution: **(a)** Let X be a subspace of $\mathbb{R}$ in the finite complement topology, and let $\mathcal{A}$ be an open cover of X. Fix $A_0 \in \mathcal{A}$. Then $X \setminus A_0$ is finite. Then for every point of $x \in X \setminus A_0$, we may pick one set $A \in \mathcal{A}$ which covers that point. The collection of these sets and A_0 is a finite subcollection of $\mathcal{A}$ which covers X.

(b) $[0,1]$ is not compact in this topology. Let $K = \left\{\frac{1}{n} \mid n \in \mathbb{Z}_+\right\}$. Let $\mathcal{A}$ be the collection of sets A_i where $A_0 = [0,1] \setminus K$ and $A_i = A_0 \cup \{1, \dots, i\}$ for $i > 0$. Then $\mathcal{A}$ is an open cover of $[0,1]$ with no finite subcover. $\blacksquare$

Problem 3

Show that a finite union of compact subspaces of X is compact.

Solution: Let $A_1, \ldots, A_n$ be compact subspaces of a space X. Suppose $\mathcal{A}$ is an open cover of $A = \bigcup_{i=1}^{n} A_i$. Then $\mathcal{A}$ is also an open cover of A_i for each i. Since each A_i is compact, there is a finite $\mathcal{A}_i \subset \mathcal{A}$ which covers A_i. Then $\bigcup_{i=1}^{n} \mathcal{A}_i$ is a finite subcover of A. ■

Problem 4

Show that every compact subspace of a metric space is bounded in that metric and is closed. Find a metric space in which not every closed bounded subspace is compact.

Solution: Suppose A is a subset of the metric space X which is not bounded.Let $x_0 \in X$. Then $\mathcal{B} = \{B_n(x_0) \mid n \in \mathbb{Z}_+\}$, the collection of open balls with integer radius centered at x_0, is an open cover of A with no finite subcover, so that A cannot be compact. Thus every compact set in X is bounded. Since X is a metric space, X is Hausdorff, and so by Theorem 26.3 every compact subspace is closed in X. Now consider $\mathbb{R}$ in the discrete topology (which is a metric space under the discrete metric). Then every subspace is closed and bounded, but only the finite subspaces are compact in this topology. ■

Problem 5

Let A and B be disjoint compact subspaces of the Hausdorff space X. Show that there exist disjoint open sets U and V containing A and B, respectively.

Solution: By Lemma 26.4, for each $x \in X$ we may find disjoint open sets U_x and V_x which contain x and Y respectively. Then the collection of all such U_x is an open cover of X, so there is a finite subcover, denote these sets by $U_1, \ldots, U_n$ (and their corresponding neighborhoods of Y by $V_1, \ldots, V_n$). Then $\bigcup_{i=1}^{n} U_i$ is an open set containing X which is disjoint from $\bigcap_{i=1}^{n} V_i$, which is an open set containing Y. ■

Problem 6

Show that if $f : X \to Y$ is continuous, where X is compact and Y is Hausdorff, then f is a closed map (that is, f carries closed sets to closed sets).

Solution: If $A \subset X$ is closed, then A is compact by Theorem 26.2, so that $f(A) \subset Y$ is compact, and thus closed by Theorem 26.3. ■

Problem 7

Show that if Y is compact, then the projection $\pi_1 : X \times Y \to X$ is a closed map.

Solution: Let $A \in X \times Y$ be closed and consider a point $x \in X - \pi_1(A)$. Since $x \notin \pi_1(A)$, $\{x\} \times Y \subset (X \times Y) - A$. Since Y is Hausdorff and $(X \times Y) - A$ is open, we may apply Lemma 26.8 and find a neighborhood W of x such that $W \times Y \subset (X \times Y) - A$. Then $x \in W \subset X - \pi_1(A)$, so $X - \pi_1(A)$ is open, so $\pi_1(A)$ is closed as desired. ■

Problem 8

Let X be a metric space with metric d; let $A \subset X$ be nonempty.

(a) Show that $d(x, A) = 0$ if and only if $x \in \bar{A}$.

(b) Show that if A is compact, $d(x, A) = d(x, a)$ for some $a \in A$.

(c) Define the ϵ-neighborhood of A in X to be the set

$$U(A, \epsilon) = \{x \mid d(x, A) < \epsilon\}$$

Show that $U(A, \epsilon)$ equals the union of the open balls $B_d(a, \epsilon)$ for $a \in A$.

(d) Assume that A is compact; let U be an open set containing A. Show that some ϵ-neighborhood of A is contained in U.

(e) Show the result in **(d)** need not hold if A is closed but not compact.

Solution: **(a)** We note that $x \notin \bar{A}$ if and only if x has a neighborhood disjoint from A if and only if there exists $\varepsilon > 0$ such that $A \cap B_d(x, \varepsilon) = \varnothing$ if and only if $d(x, a) > \varepsilon$ for this ε for all $a \in A$ if and only if $d(x, A) \geq \varepsilon > 0$.

(b) Define $f : A \to \mathbb{R}$ by $f(a) = d(x, a)$. Since A is compact, we have $a \in A$ where f attains its minimum. This is a such that $f(a) = d(x, a) = d(x, A)$.

(c) If $x \in B_d(a, \varepsilon)$ for some $a \in A$ and $\varepsilon > 0$, then $d(x, A) \leq d(x, a) < \varepsilon$, so $x \in U(A, \varepsilon)$. Suppose $x \in U(A, \varepsilon)$. Then $d(x, A) = \inf_{a \in A} d(x, a) < \varepsilon$, so there exists $a \in A$ such that $d(x, a) < \varepsilon$ (otherwise $d(x, A) \geq \varepsilon$). Hence $x \in B_d(a, \varepsilon)$.

(d) Define $f : A \to \mathbb{R}$ by letting $f(a) = d(A, X - U)$. Let δ be the minimum value taken by f. Let $a \in A$ such that $f(a) = \delta$. Since $a \in U$, there is $\varepsilon > 0$ such that $B_d(a, \varepsilon) \subset U$, so $\delta = f(a) > \varepsilon > 0$. Hence $U(A, \delta) \subset U$ with $\delta > 0$.

(e) Let $A = \{x \times y \mid 0 < x = 1/y\}$. Let $U = \{x \times y \mid 0 < x < 1/(2y)\}$. Then $B_d(x \times y, \min\{x/2, y/2\}) \subset U$, but $d(x \times 1/x, x \times 1/(2x)) \leq 1/(2x)$, which grows arbitrarily small for large x. ■

Problem 9

Show that a connected metric space having more than one point is uncountable.

Solution: Suppose our metric space (X, d) with at least two points is countable. Let $x, y \in X$ with $d(x, y) > 0$. Since $[0, d(x, y)]$ is uncountable, there exists $0 < r < d(x, y)$ such that $d(x, z) \neq r$ for all $z \in X$. Since the closure of $U = B_d(x, r)$ is a subset of the closed ball of radius r, the complement V of this closure is open and contains all points $z \in X$ with $d(x, z) > r$. Notice that $U \cap V = \varnothing$ and $U \cup V = X$, as $d(x, z) < 0$ or $d(x, z) > 0$ for each $z \in X$, but not both. Since $x \in U$ and $y \in V$, we have that U and V are a separation of X, our countable space with at least two points. ■

Problem 10

Let X be limit point compact.

(a) If $f : X \to Y$ is continuous, does it follow that $f(X)$ is limit point compact?

(b) If A is a closed subset of X, does it follow that A is limit point compact?

(c) If X is a subspace of the Hausdorff space Z, does it follow that X is closed in Z ?

We comment that it is not in general true that the product of two limit point compact spaces is limit point compact, even if the Hausdorff condition is assumed. But the examples are fairly sophisticated.

Solution: **(a)** No. Consider the set X of Example 1. Then X is limit point compact but the projection map π_1 is continuous and maps X onto $\mathbb{Z}_+$, which is not limit point compact.

(b) Yes. Let $B \subset A$ be infinite. Then B has a limit point in X. But this is a limit point of A. Since A is closed, the limit point belongs to A, so B has a limit point in A.

(c) No. Consider $S_\Omega \subset \bar{S}_\Omega$. Then S_Ω is not closed but it is limit point compact. ■

Problem 11

Let (X, d) be a metric space. If $f : X \to X$ satisfies the condition

$$d(f(x), f(y)) = d(x, y)$$

for all $x, y \in X$, then f is called an isometry of X. Show that if f is an isometry and X is compact, then f is bijective and hence a homeomorphism. **Hint:** If $a \notin f(X)$, choose ϵ so that the ϵ-neighborhood of a is disjoint from $f(X)$. Set $x_1 = a$, and $x_{n+1} = f(x_n)$ in general. Show that $d(x_n, x_m) \geq \epsilon$ for $n \neq m$.

Solution: The continuity of f (and its inverse, which we will show exists) follow easily from the fact that $d(x,y) < \epsilon \Longleftrightarrow d(f(x), f(y)) < \epsilon$. Suppose $x_1, x_2 \in X$ and $f(x_1) = f(x_2)$. Then

$$0 = d(f(x_1), f(x_1)) = d(f(x_1), f(x_2)) = d(x_1, x_2)$$

so that $x_1 = x_2$, so f is injective. Now assume f is not surjective. Let $a \in X - f(X)$. Since X is compact, $f(X)$ is compact, and hence closed, in X. So we may fix $\epsilon > 0$ so that $B(a, \epsilon)$ is disjoint from $f(X)$. Define $x_0 = a$ and $x_{n+1} = f(x_n)$ for $n \in \mathbb{Z}_+$. Then $x_n \in f(X)$ for each $n \in \mathbb{Z}_+$. Then $d(x_n, x_m) = d(a, x_{n-m}) \geq \epsilon$ for $n > m$. Let $x \in X$. Then $B(x, \epsilon)$ contains at most one x_i different from x, so we may find a neighborhood of x which does not contain any of the points x_i, except possibly x, so x is not a limit point of the infinite set $A = \{x_n \mid n \in \mathbb{Z}_+\}$. Nut then A has no limit points, a contradiction to the limit point compactness of X. Thus f is surjective, and hence bijective. ■

Problem 12

On $\mathbb{R}^{n+1} \backslash \{0\}$ define the equivalence relation

$$x \sim y \Leftrightarrow \text{there is a nonzero real constant } \lambda \text{ such that } x = \lambda y.$$

The real projective n-space is defined by $\mathbb{RP}^n = (\mathbb{R}^{n+1} \backslash \{0\}) / \sim$. Show that $\mathbb{RP}^n$ is a manifold.
Hint: build charts (U_i, φ_i) with

$$U_i = \{[x_1, ..., x_{n+1}] \in \mathbb{RP}^n \mid x_i \neq 0\},$$

$1 \leq i \leq n+1$.

Solution: We consider the map given by $\varphi_i : U_i \to \mathbb{R}^n$

$$\varphi_i([x_1, ..., x_{n+1}]) = \left(\frac{x_1}{x_i}, ..., \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, ..., \frac{x_{n+1}}{x_i} \right).$$

Since $x_i \neq 0$, it is clearly well-defined. It is clearly a bijection, and its image $\varphi_i(U_i) = \mathbb{R}^n$ is open. The inverse is given by

$$\varphi_i^{-1}(u_1, ..., u_n) = [u_1, ..., u_{i-1}, 1, u_{i+1}, ..., u_n].$$

The union of the domains U_i clearly covers $\mathbb{RP}^n$. So, it remains to show that the overlap maps are smooth diffeomorphisms. They are given by

$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$$

with, for $j < i$ (this is similar for $j > i$),

$$\varphi_j \circ \varphi_i^{-1}(u_1, ..., u_n) = \left(\frac{u_1}{u_j}, ..., \frac{u_{j-1}}{u_j}, \frac{u_{j+1}}{u_j}, ..., \frac{u_{i-1}}{u_j}, \frac{1}{u_j}, \frac{u_{i+1}}{u_j}, ..., \frac{u_n}{u_j} \right)$$

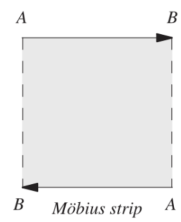
where

$$\varphi_i(U_i \cap U_j) = \{(u_1, ..., u_n) \in \mathbb{R}^n \mid u_j \neq 0\}$$
$$\varphi_j(U_i \cap U_j) = \{(u_1, ..., u_n) \in \mathbb{R}^n \mid u_i \neq 0\}.$$

The maps $\varphi_j \circ \varphi_i^{-1}$ are clearly smooth diffeomorphisms.

Problem 13

The Möbius strip can be constructed as the quotient of $(0, 1) \times [0, 1]$ by the equivalence relation determined by $(x, 0) \sim (1 - x, 1)$, see the following figure.



Prove that the Möbius strip is a manifold, by showing that the following two maps provide an atlas:

$$\varphi_1 :]0, 1[\times]0, 1[\to]0, 1[\times]0, 1[\,, \quad \varphi_1(x, y) = (x, y)$$
$$\varphi_2 : \big(]0, 1[\times\big([0, \epsilon[\cup]1 - \epsilon, 1]\big)\big)/\sim \to]0, 1[\times] - \epsilon, \epsilon[,$$
$$\varphi_2([x, y]) = \begin{cases} (x, y) & \text{if } y \in [0, \epsilon[\\ (1 - x, y - 1) & \text{if } y \in]1 - \epsilon, 1] \end{cases}$$

Solution These maps are clearly bijections onto open subsets of $\mathbb{R}^2$ and their domain cover the Möbius strip. The transition map is given by

$$\begin{aligned} \varphi_1 \circ \varphi_2^{-1} : \varphi_2(U_1 \cap U_2) &=]0, 1[\, \times \big(\,] - \epsilon, 0[\cup]0, \epsilon[\, \big) \\ \to \varphi_1(U_1 \cap U_2) &=]0, 1[\, \times \big(\,]0, \epsilon[\cup]1 - \epsilon, 1[\, \big) \end{aligned}$$

$$\varphi_1 \circ \varphi_2^{-1}(u_1, u_2) = \begin{cases} (u_1, u_2) & \text{if } u_2 \in]0, \epsilon[\\ (1 - u_1, u_2 + 1) & \text{if } u_2 \in] - \epsilon, 0[. \end{cases}$$

It is a C^∞ diffeomorphism between open (non connected) subsets of $\mathbb{R}^2$. Therefore, $\mathcal{A} = \{(U_1, \varphi_1), (U_2, \varphi_2)\}$ is an atlas.